

$$13.7) \quad u_o(t) = u_o(t_0) - \frac{u_i}{50 \times 10^3 \times 0.02 \times 10^{-6}} (t - t_0) \\ = u_o(t_0) - 1000 u_i (t - t_0)$$

i) $t = 0 \sim 1 \text{ ms}$, $u_i = 5 \text{ V}$

$$u_o(t) = u_o(0) - 1000(5)(t - 0) \\ = -5000t$$

$$u_o(0) = 0 \quad u_o(1 \text{ ms}) = -5 \text{ V}$$

ii) $t = 1 \sim 3 \text{ ms}$, $u_i = -3 \text{ V}$

$$u_o(t) = u_o(1 \text{ ms}) + 3000(t - 1 \text{ ms})$$

$$u_o(3 \text{ ms}) = -5 + 3000(2 \text{ ms})$$

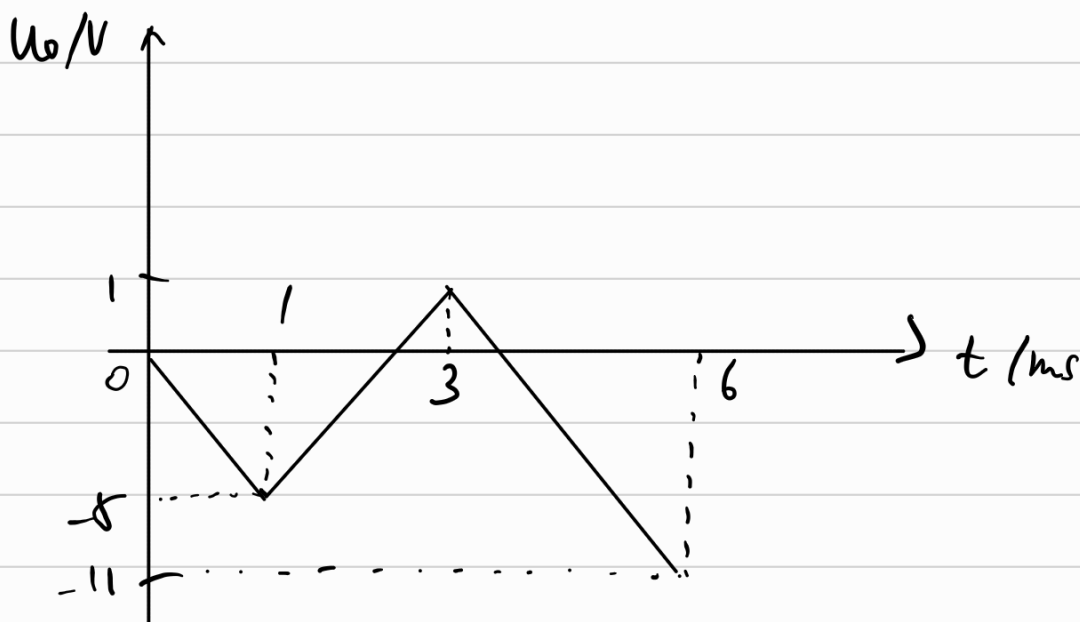
$$u_o(3 \text{ ms}) = 1 \text{ V}$$

iii) $t = 3 \sim 6 \text{ ms}$, $u_i = 4 \text{ V}$

$$u_o(t) = u_o(3 \text{ ms}) - 1000(4)(t - 3)$$

$$u_o(6 \text{ ms}) = 1 - 4000(3 \text{ ms}) = -11 \text{ V}$$

$$u_o(t) = \begin{cases} -5000t, & 0 \leq t \leq 1 \text{ ms} \\ -5 + 3000(t - 1 \times 10^{-3}), & 1 \leq t \leq 3 \text{ ms} \\ 1 - 4000(t - 3 \times 10^{-3}), & 3 \leq t \leq 6 \text{ ms} \end{cases}$$



$$13.8) T = RC = 8 \times 10^3 \times 0.001 \times 10^{-6} = 8 \mu s$$

$$u_o = -RC \frac{du_i}{dt} = -8 \times 10^{-6} \frac{du_i}{dt}$$

i) $t = 0 \sim 5 \mu s$

$$t = 0 \mu s, u_i = 0 V, t = 5 \mu s, u_i = 5 V$$

$$u_i = 1 \times 10^6 t \text{ (V)}$$

$$u_o = -8 \times 10^{-6} \frac{du_i}{dt} = -8 \times 10^{-6} \times 1 \times 10^6 = -8 V$$

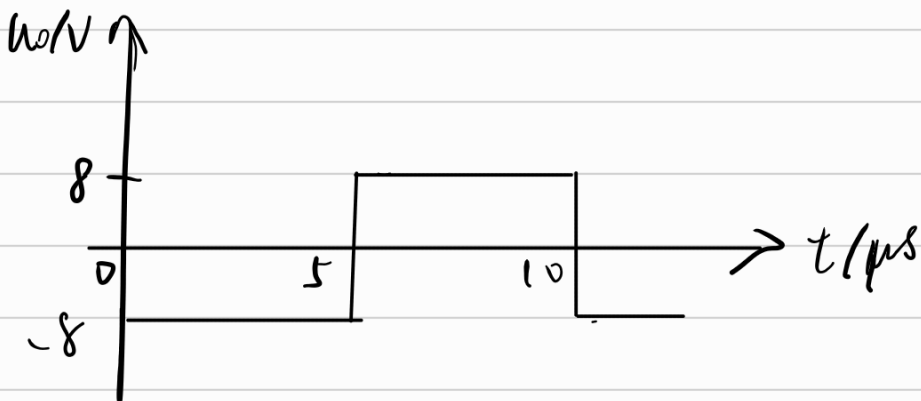
ii) $t = 5 \sim 10 \mu s$

$$t = 5 \mu s, u_i = 5 V, t = 10 \mu s, u_i = 0 V$$

$$u_i = -1 \times 10^6 (t - 5 \times 10^{-6}) + 5$$

$$= -1 \times 10^6 t + 10$$

$$u_o = -8 \times 10^{-6} (-1 \times 10^6) = 8 V$$



$$1318) a) u_{01} = -\frac{R}{2R} u_{i1} = -\frac{1}{2} u_{i1}$$

$$u_{12}' = \frac{1}{2} (u_{01} + u_{i2}) = -\frac{1}{4} u_{i1} + \frac{1}{2} u_{i2}$$

$$R' = R \parallel R = \frac{R}{2}$$

$$u_0 = \frac{5R}{\frac{R}{2}} (u_{i3} - u_{12}') \\ = 10 (u_{i3} + \frac{1}{4} u_{i1} - \frac{1}{2} u_{i2})$$

$$b) u_{01} = u_1 \\ u_{02} = u_2 \\ u_{03} = u_3$$

$$V = \frac{\frac{u_1}{R} + \frac{u_2}{2R} + \frac{u_3}{3R}}{\frac{1}{R} + \frac{1}{2R} + \frac{1}{3R}} = \frac{6}{11} (u_1 + \frac{u_2}{2} + \frac{u_3}{3})$$